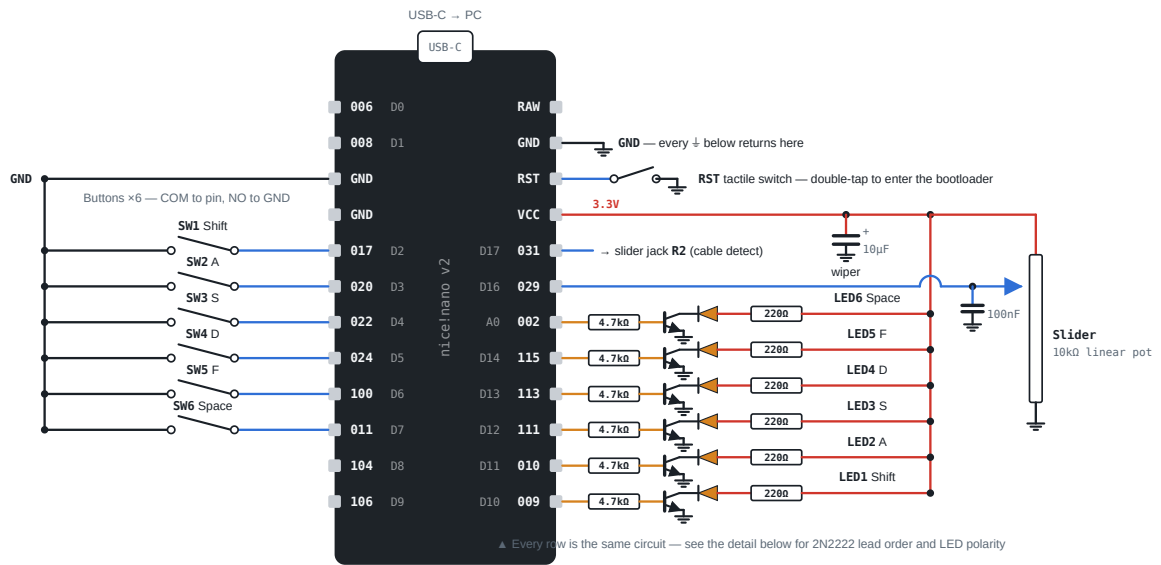


A. Base wiring — no LEDs (buttons + slider)



B. With LEDs — buttons + slider + 6 LEDs

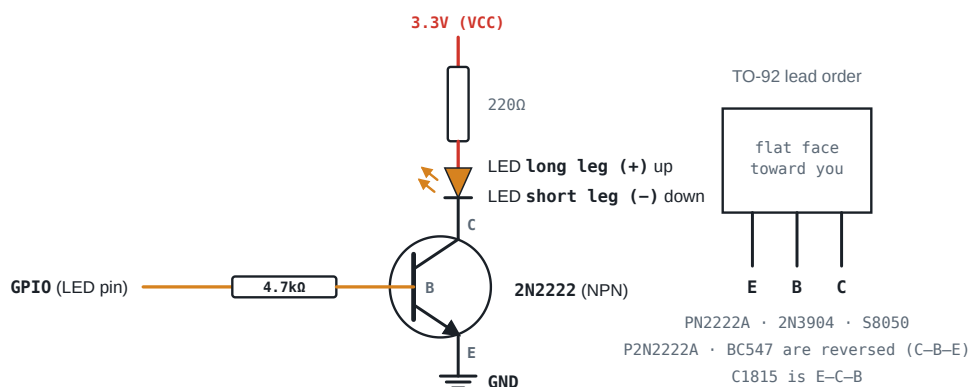


Board seen from the front, USB-C at the top. A dot is a junction, a hop is a wire crossing without connecting, $\perp$ is GND (same net as the board GND pin). Large numbers on the board are the silkscreen labels; small ones are the Arduino pin numbers used in code.

— Input (buttons · slider → MCU) — Output (MCU → transistor) — 3.3V — GND

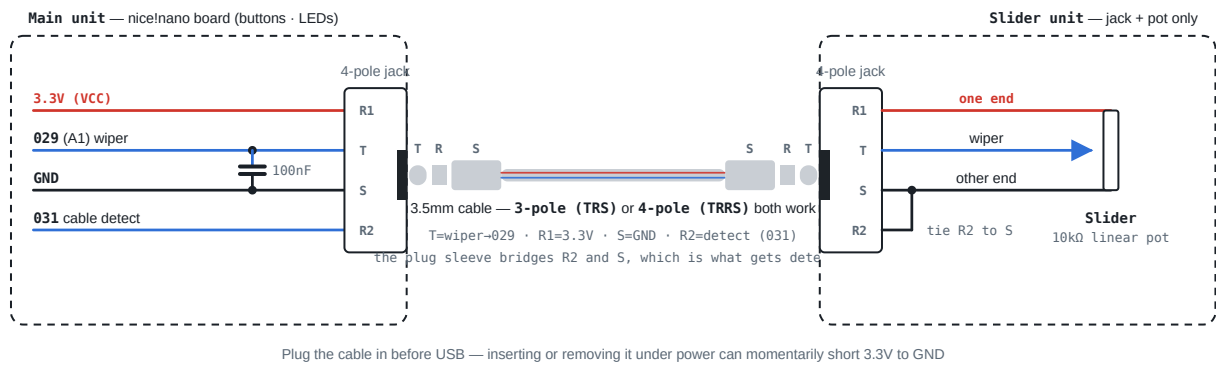
⚡ ground symbol = same GND net

LED driver cell (×6, all identical)



When the GPIO goes HIGH, 0.55mA flows into the base and the transistor turns on; about 5mA then flows through the LED from 3.3V via 220Ω. LED current never passes through the MCU pin, so lighting all six at once stays clear of the GPIO limits.

Detachable slider unit — 3.5mm 4-pole (TRRS) jack



Main-unit jack: T→029, R1→3.3V, S→GND, R2→031 (detect). Slider-unit jack: T→wiper, R1·S→the two ends, R2 tied to S. When a cable is inserted, 031 on the main unit drops to GND and the firmware registers the connection (LEDs flash once); on removal it stops sending mouse reports (LEDs flash twice). The 100nF lives on the main unit so it also filters noise picked up by the cable. The slider ends can be swapped; if the direction is backwards, flip `MOUSE_INVERT_X`.

Pin assignment

Part	Board label	Code pin	Connection	Key / role
SW1	017	D2	button → GND	Shift
SW2	020	D3	button → GND	A
SW3	022	D4	button → GND	S
SW4	024	D5	button → GND	D
SW5	100	D6	button → GND	F
SW6	011	D7	button → GND	Space
RST switch	RST	—	tactile switch → GND	Reset · double-tap for bootloader
LED1	009	D10	4.7kΩ → 2N2222 B	Shift indicator
LED2	010	D11	4.7kΩ → 2N2222 B	A indicator
LED3	111	D12	4.7kΩ → 2N2222 B	S indicator
LED4	113	D13	4.7kΩ → 2N2222 B	D indicator

Part	Board label	Code pin	Connection	Key / role
LED5	115	D14	4.7kΩ → 2N2222 B	F indicator
LED6	002	D15	4.7kΩ → 2N2222 B	Space indicator
Slider	029	A1	jack T ← wiper · 100nF → GND	Mouse X
Cable detect	031	D17	jack R2 (GND when plugged, internal pull-up)	Slider presence
3.3V	VCC	—	slider one end · 220Ω ×6 · 10μF (+)	Power
GND	GND	—	buttons ×6 · slider other end · 2N2222 E ×6 · capacitor —	Ground
Unused	006 008 104 106	—	leave open	RAW and RST also unused

Bill of materials

Items marked (Korean) were bought from a Korean shop. If you cannot source them locally, importing via the AliExpress links or search terms is the better option. **Button lighting is optional** — the LED, 2N2222, 220Ω and 4.7kΩ are only needed if you fit LEDs. The six LED driver cells can be built on a **separate board** and mounted in free space inside the case (see notes).

Part	Needed	Purpose · what to look for	Buy
nRF52840 Pro Micro-compatible board	1	nice!nano v2 or any clone with the same pinout (SuperMini NRF52840 etc.). USB-C; check the photos that the pin labels (009, 111 ...) match this page	AliExpress

Part	Needed	Purpose · what to look for	Buy
Perfboard 4×6cm double-sided	1	Only 4×6cm fits the case. Double-sided plated-through board. Check the corner M3 holes line up with the case bosses	AliExpress
Slide potentiometer B10K	1	B (linear) 10kΩ. A (log) will not work. 60mm travel — a 100mm pot does not fit the housing	AliExpress
3.5mm 4-pole (TRRS) jack breakout PJ-320A	2	1 main unit · 1 slider unit. Needs all four pins T/R1/R2/S broken out	AliExpress
CWB 405 button (includes LED holder · 60g micro switch · 20g spring; LED not included)	6	The button body contains a 5mm LED holder (side locking-pin type), a 60g micro switch and a 20g spring. LED sold separately. For a better feel, swap in the Omron switch below	IST Mall (Korean) eBay (same part, ships worldwide)

Part	Needed	Purpose · what to look for	Buy
Micro switch (pin plunger)	6	Optional — the 60g switch in the 405 button is fine; swap in an Omron pin plunger only if you want a better feel (Korean shop). Outside Korea, VX-5-1A3 (50g, solder terminals) or equivalent. SPDT or SPST-NO both fine (only COM and NO are used). 25g / 50g actuation to taste	IST Mall (Korean) AliExpress
Tactile switch 6×6mm (for RST)	1	Between RST and GND. Double-tap to enter bootloader mode for firmware upload	search: 6x6 tactile switch
LED 5mm (5V)	6	(For LED, optional) As built: 5V LED (Korean shop). Any colour; diffused reads better off-axis. If it has a built-in resistor (5V-direct type), replace the 220Ω with a jumper	IST Mall (Korean) search: 5mm LED diffused

Part	Needed	Purpose · what to look for	Buy
2N2222 (NPN transistor, TO-92)	6	(For LED, optional) LED driver. 2N3904 · S8050 · BC547 also work (check lead order)	search: 2N2222 TO-92
220Ω resistor 1/4W	6	(For LED, optional) LED current limit (~5mA)	search: resistor kit 1/4W
4.7kΩ resistor 1/4W	6	(For LED, optional) Transistor base current limit	
100nF (104) ceramic	1	Slider signal filter	search: 104 ceramic capacitor
10μF / 16V electrolytic	1	3.3V rail decoupling	search: 10uF 16V electrolytic
3.5mm cable, male-male	1	Main unit ↔ slider unit. 3-pole (TRS) or 4-pole (TRRS). 1m recommended	search: 3.5mm aux cable 1m
Pin header 2.54mm 1×12	2	Mounts the board above the perfboard. Often included with the board	search: pin header 2.54mm 40pin
M2 × 5mm screw	8	Perfboard mounting	search: M2 5mm screw
M3 × 10mm screw	14	Case bottom assembly	search: M3 10mm screw

Part	Needed	Purpose · what to look for	Buy
Wrapping wire 30AWG	–	Back-side wiring. At least 3 colours (GND black · 3.3V red · signal)	search: 30AWG wrapping wire

Things to watch

Transistor lead order

Same TO-92 shape, different lead order from part to part. Read the marking on the body and match it to the table in the detail above. If unsure, use a multimeter in diode mode: red lead on one leg — if both other legs read 0.6–0.7V, that leg is B.

Swapping C and E does no damage, but the LED will be very dim or off. If you see that, rotate the transistor 180°.

The two resistors do different jobs

220Ω sets the LED current ($(3.3V - 2V) \div 220 \approx 5mA$). **4.7kΩ** limits the GPIO-to-base current to 0.55mA and protects the nRF pin. Neither can be omitted and they cannot be merged. LED current comes from the 3.3V regulator, so all six lit at once (30mA) is fine. The GPIOs together only source 3.3mA.

LED polarity

Long leg (+, anode) toward the 220Ω, short leg (–, cathode) toward the transistor C. Backwards it just stays off — no damage.

Brightness is adjustable in firmware via PWM; the 220Ω sets the maximum.

The two capacitors

100nF between the slider wiper (029) and GND, **close to the board**, so it also filters noise picked up by the wiring. No polarity.

10μF between VCC and GND on the board that carries the LEDs and slider. It absorbs the 3.3V dip when LEDs switch on. For an electrolytic, long leg (+) to VCC, white stripe to GND.

Buttons (micro switches)

No polarity. Terminal **COM** to the pin, **NO** to GND. On a 3-terminal (SPDT) switch **leave NC open** — wiring NC instead reads as pressed when idle. The pull-up is inside the chip, so no external resistor.

If the terminals are unmarked, use continuity mode: the pair that beeps at rest is COM-NC, the pair that beeps when pressed is COM-NO.

RST switch

One small tactile switch between RST and GND. A single press reboots; a **quick double-tap** enters bootloader mode (a NICENANO drive appears) so you can flash firmware. Place it where it can be reached from outside the case.

Perfboard too small? Split off the LED circuit

Important. The six LED driver cells (transistors and resistors) do not have to live on the main board. If 4×6cm is getting tight, solder just the six LED cells onto a separate small perfboard (e.g. 3×7cm or an offcut) and fix it in empty space inside the case with double-sided tape or hot glue.

Only **8 wires** need to cross between the boards: the six GPIOs (009 · 010 · 111 · 113 · 115 · 002) plus 3.3V and GND. Put the 10µF on the LED board.

Identifying 4-pole jack terminals

Which terminal is T/R1/R2/S varies by product. Insert a **3-pole** cable, then use continuity mode to match each terminal to the tip, ring and sleeve on the far end of the plug. **Two terminals will beep with the sleeve**: the one bonded to the metal body is S, the other is R2.

PJ-320A breakouts usually have T/R1/R2/S or 1-4 printed on the silkscreen.

Slider

The two end pins go to VCC (3.3V) and GND, the wiper to **029**. If the direction is backwards, swap the ends or flip `MOUSE_INVERT_X` in code. Use VCC, not RAW, or the ADC range will be off.